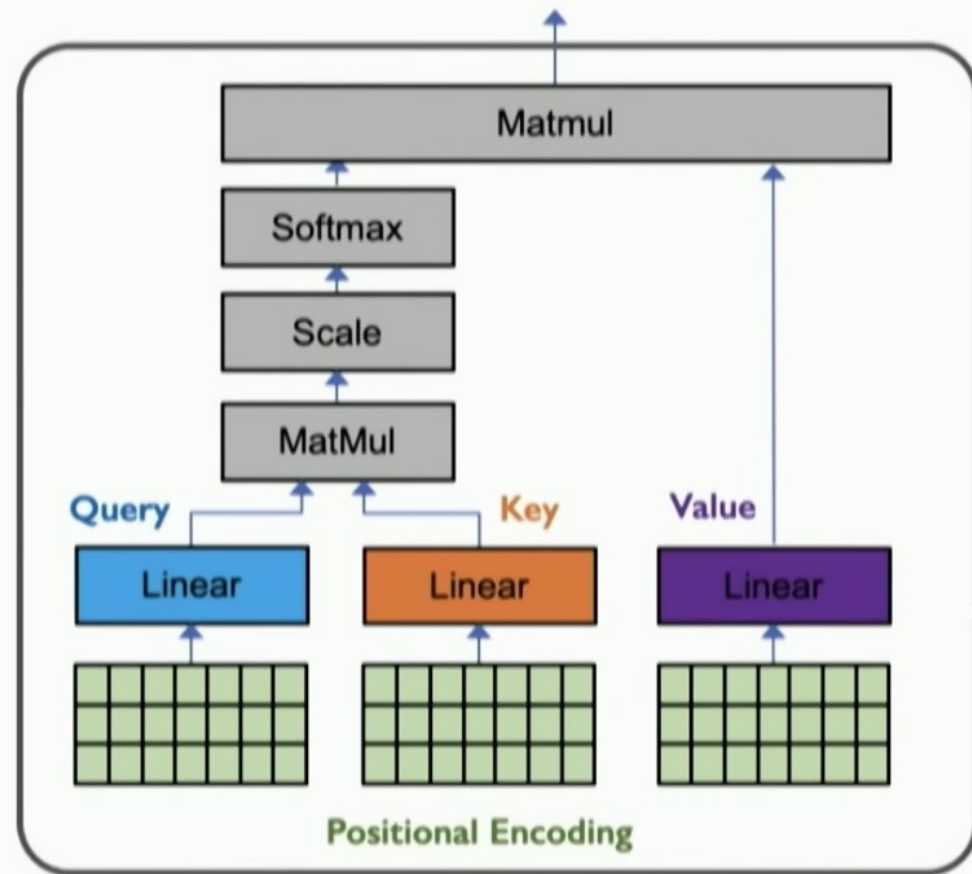


The attention matrix (the softmax output) is then used to weight the Value (V) vectors.

This is done by multiplying the attention matrix with the Value matrix.

This means that the vectors/columns of the Value matrix are “weighted” through the scores of the attention matrix.
This is the output of the attention layer.

1. Encoding the **position**
2. Extracting **query, key, value**
3. Compute **attention weighting**
4. Extract **features with high attention**



$$\text{softmax} \left(\frac{Q \cdot K^T}{\text{scaling}} \right) \cdot V$$